

Answers for Falling Cow Project, Computational Physics

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Is energy conserved as a function of time? To what level of precision?

Yes, when there's no drag, the total energy stays basically constant. In our runs with $dt = 0.001$, it only changed by about 1 part in 10,000 from start to finish. It's not perfectly conserved because the code takes finite time steps, and each step has a tiny error that builds up over time. The kinetic and potential energy curves are mirror images of each other, and their sum stays flat.

How does the trajectory change if you change the time step? What about as dt approaches 0?

The bigger the time step, the more the trajectory drifts away from the true answer, and the more the total energy changes over time. When we made dt smaller ($0.1 \rightarrow 0.01 \rightarrow 0.001 \rightarrow 0.0001 \rightarrow 0.00001$), the trajectory got closer and closer to the analytic way, and the energy line got flatter. As dt goes toward zero, the numerical answer approaches the exact solution. But you can't actually use $dt = 0$ since the code would never finish, and there's still round-off error from the computer's math we guess.

How does your trajectory compare to the analytic solution if you set air resistance to zero?

They match. We plotted the numerical trajectory and the analytic prediction on the same graph as you can see and as you can test with the code, the two curves sit right on top of each other. So the code is doing the physics correctly.

Drag free case:

Do the numerical and analytic trajectories agree?

Yes, at $dt = 0.001$, they overlap perfectly on the plot.

Is the total energy conserved?

Yes, to within a very small numerical error. The KE and PE swap back and forth while their sum stays flat.

What pattern did you observe when changing dt ?

Smaller dt = more accurate trajectory and flatter energy. Each time we made dt $10\times$ smaller, the energy drift got about $10\times$ smaller too. At $dt = 0.1$ the drift is obvious, at $dt = 0.001$ it's already good, and at $dt = 0.00001$ it doesn't get much better but the code takes way longer to run. So $dt = 0.001$ is a good middle ground.

Air resistance case (drag = 4):

Are there conserved quantities?

No, energy is not conserved anymore. Drag takes energy away from the cow (it becomes heat in the air), so on the energy plot the total energy just keeps going down.

Are there any important limits?

Yes. When the drag force gets big enough to cancel out gravity, the cow stops speeding up and falls at a constant speed, which our simulation is showing that too.

What does the trajectory look like?

It's not a nice parabola anymore. The cow doesn't travel as far horizontally because drag slows down v_x too, and the fall takes longer because the cow can't go faster than terminal velocity. The path curves down more steeply at the end.